



A self-calibration method for capacitance mismatch in SAR ADC with split-capacitor DAC

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Abstract

A self-calibration method to calibrate the nonlinearity due to capacitance mismatch in successive approximation register (SAR) analog-to-digital converter (ADC) is presented. It focuses on calibrating the most significant bit (MSB) array in the split-capacitor main DAC (split-MDAC) by using a calibration DAC (CDAC) that contains multiple sub-CDACs. Every bit in MSB array has its corresponding sub-CDAC in CDAC, which enhances the calibration efficiency. To verify the calibration method, a 14 bit, 500kS/s SAR ADC is implemented, and it is manufactured in 0.35 μ m 2P4M CMOS process. The measured results show that the proposed calibration method can assist this SAR ADC to achieve better static and dynamic performance, and its ENOB is improved from 9 bit to 11.98 bit at Nyquist input frequency.

Introduction

Charge-Redistribution (CR) successive approximation register (SAR) analog-to-digital converter (ADC) is widely used in portable devices and high-density integration devices due to its lowest power consumption [1], [2]. However, high performance applications require low power SAR ADC with high resolution (e.g., 12–16 bit).

In the traditional SAR ADC with binary weighted capacitor DAC, a high-resolution SAR ADC usually contains a large capacitor array, which limits the ADC dynamic performance seriously. To improve the resolution and area efficiency, lots of efforts have been done to reduce the array area. These efforts can be classified into two strategies. One is decreasing the size of unit capacitor [3] and the other is improving the architecture of DAC by decreasing the amount of unit capacitors. However, the size of unit capacitor is limited by thermal noise and power noise which make it hard to be reduced further. Therefore, the attractive choice to reduce the DAC area overhead is decreasing the amount of unit capacitors. The common method to decrease the number of capacitors is dividing the whole array into several segments by bridge capacitors, namely split-capacitor main DAC (split-MDAC) [4], [5], [6], [7], [8], [9], [10], [11], [12], [13]. These split structures are sensitive to two factors, namely parasitic capacitors around bridge capacitors and capacitance mismatches caused by process variations. Both these two factors can make the nonlinearity distortion even more worse [8], [9], [10], especially when the unit capacitance is cut down. Thus, in order to achieve a high resolution with acceptable area overhead, split structure should be chosen, and the effect of parasitic capacitor and capacitance mismatch should be considered.

To compensate the nonlinearity caused by parasitic capacitances around bridge capacitor, a tunable capacitor in parallel with the least significant bit (LSB) array is used [8], [9]. The tunable capacitor can balance the weight around bridge capacitor to make the LSB capacitance of MSB array equal to the capacitance seen from the MSB array toward the LSB array through the bridge capacitor. To calibrate the capacitance mismatches caused by process variations, dynamic calibration is required as every calibrated bit should be calibrated respectively. Works [14], [15] first propose a self-calibration technique to calibrate the mismatch, which improves its accuracy without a factory calibration or a golden standard [16], yet its DAC is a hybrid structure without split structure. The calibration method mentioned in [14] is implemented in [9] with an extra calibration DAC to measure and compensate the nonlinearity of the C_{M-1} three bits. Work [5] is also based on the method in [14] and illustrates the detailed calibration operating. But it needs a negative reference voltage to calibrate the negative mismatch, which makes the system more complex. In this work, a self-calibration method for capacitance mismatch is presented which is based on the method in [14], but this work extends it in following aspects:

1. Developing the calibration method into split-MDAC architecture which is more popular than traditional binary weighted MDAC, especially for high resolution ADC.
2. Taking the negative error voltage into account without involving a negative reference voltage.
3. Using multiple sub-calibration DACs to enhance the calibration efficiency.

It focuses on calibrating the mismatches of most significant bit (MSB) array in MDAC which have greater influence on linearity than that of least significant bit (LSB) array [17]. Together with the

proposed calibration DAC (CDAC), this self-calibration method can be used to calibrate the split-MDAC with two or more sub-DACs.

The rest of the paper is organized as follows: Section 2 introduces ADC architecture with proposed calibration technique. The detailed implementation of the calibration technique is illustrated in three main stages in 3 The implementation of calibration method, 4 Application example and results presents the application example with the proposed calibration technique and the results. Finally, Section 5 concludes this paper.

Section snippets

The architecture of ADC with proposed calibration method

The three most common split-MDAC structures are presented in Fig. 1. Fig. 1(a) shows a structure which uses all the capacitors to sample input voltage (V_{in}). It requires a fractional-valued bridge capacitor to guarantee the value of any bit capacitor to be equal to the sum of all lower bit capacitors [11], [12], [13]. Fig. 1(b) shows a structure which only uses the MSB array to sample V_{in} . It uses an unit capacitor as the bridge capacitor so that the fractional-valued bridge capacitor is ...

The implementation of calibration method

Fig. 3 illustrates the operating timing of the whole process. After powered-on, ADC first jumps into the *CALIB_C_p* step to calibrate the parasitic capacitors around bridge capacitor [8]. Then, it begins to calibrate the capacitor mismatch by three main stages, namely *Getting calibration code (GCC)*, *Sampling & holding (S&H)*, and *Conversion&calibration (C &C)*. Every stage consists of several sub-stages as shown in Fig. 3. The following subsections will describe every sub-stage in detail. The ...

Application example and results

To verify the proposed calibration method, a 14 bit, 500 kS/s SAR ADC is implemented. It contains 4+4+6 (4-bit 1st LSB array, 4-bit 2nd LSB array and 6-bit MSB array) MDAC arrays and 6 sub-CDAC arrays. This ADC is manufactured in 0.35 μ m 2P4M CMOS process with PIP capacitors. To avoid the badly influence of parasitic capacitors on metal wires, the unit capacitance in MDAC is chosen as 180 fF. As the linearity of CDAC is relatively non-significant compared with MDAC, so, the unit capacitance in ...

Conclusion

To obtain a high resolution SAR ADC, a self-calibration method has been presented. This calibration method can be implemented in the ADC with split-capacitor MDAC array. It focuses on the

calibration of MSB array since the mismatch in MSB array causes more serious effects on linearity than LSB array. To calibrate the mismatch in MSB array of MDAC, a CDAC with multiple sub-CDACs is utilized to enhance calibration efficiency. The detailed operating principle and timing of MDAC and CDAC are ...

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